

Dirac spinor normalization

A widely-used convention is

$$u^\dagger u = 2E$$

where E is relativistic energy

$$E = \sqrt{\vec{p} \cdot \vec{p} + m^2}$$

For example, for a spin-up electron

$$u = \sqrt{E + m} \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \end{pmatrix} \quad u^\dagger u = 2E \quad (1)$$

To demonstrate Lorentz invariance, let $\vec{p} = 0$. We have $\sqrt{E + m} = \sqrt{2m}$ hence

$$u = \sqrt{2m} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad u^\dagger u = 2m = 2E$$

It is instructive to see what happens when u is not normalized.

$$u = \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \end{pmatrix} \quad u^\dagger u = \frac{\vec{p} \cdot \vec{p}}{(E + m)^2} + 1 \quad (2)$$

The issue is $u^\dagger u$ is not Lorentz invariant. For example, in the rest frame $\vec{p} = 0$ we obtain

$$u^\dagger u = 1$$

Eigenmath script